EFE Structural Proofs: Curvature Encoding, Coefficient Identification, and Diffeomorphism Invariance

Discharging Obligations NFC-5, NFC-6, NFC-7

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Discharge record. This paper discharges three obligations from Paper D2:

- **NFC-5** (O-KPO3-Curv): KPO-3 encoding identification for the curvature observable F_{geom}.
- **NFC-6** (O-EFE-Coeff): Structural identification of EFE coupling constants $(\kappa, \Lambda, c_{\text{GB}})$ with NFC parameters.
- **NFC-7** (O-Diff-Inv): Diffeomorphism invariance of $g_{\mu\nu}$ from PASS.

These three obligations share the KPO-3 encoding infrastructure and are most efficiently discharged together. All three were attributed to NFC Theorems 41.15, 41.17 in Papers BH and BP. The proofs below use only suite objects already established at Conditional status.

Abstract

We provide suite-internal proofs of three EFE structural results that were previously attributed to NFC Theorems 41.15 and 41.17. §2 proves the KPO-3 encoding identification for the curvature observable (NFC-5): $F_{\text{geom}}(\sigma) = [(\lambda_2 - \lambda_1)/\lambda_1] \cdot \text{tr}(F_\sigma \cdot \Gamma(\sigma))$ arises from the response algebra structure and the KPO-3 spectral encoding. §3 proves the structural EFE coefficient identification (NFC-6): κ is the unique dimensionless coupling consistent with the NFC interaction scales; Λ is the screened-sector ground-state energy; and c_{GB} vanishes by the Chern–Gauss–Bonnet theorem in $d = 4$. §4 proves diffeomorphism invariance of $g_{\mu\nu}$ (NFC-7) from PASS screening and gauge covariance.

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1 Imports

- **AV1** (Paper AV): Gauge algebra $\mathfrak{g} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ at $K_0 = 7$. Response algebra $\mathcal{O}_\partial \subset \mathcal{A}_\infty$ of dimension $\dim \mathcal{O}_\partial = 9$. Status: Conditional (SA2+Phase-A).
- **BH-Lich** (Paper BH, using Lichnerowicz 1958 = category B in Paper D2): The covariant Laplacian L_∞ on $\mathcal{A}_\infty$ has spectral ratio $(\lambda_2 - \lambda_1)/\lambda_1 \to \mathrm{Ric}/m^2$ in the continuum limit. Status: Conditional (Phase-A+KPO-3).
- **BH-Lovelock** (Paper BH, entry BH-Thm-Lovelock, category A in Paper D2): $\dim \mathcal{O}_\partial = 9 = 1+4+4$ matches the Lovelock gravity decomposition in $d = 4$. Status: Conditional (Phase-A).
- **AZ1** (Paper AZ): KMS state ω_β on $\mathcal{A}_\infty$ at inverse temperature $\beta > 0$. Massive screened-sector ground state with ground-state energy $E_0 = m^2$. Status: Conditional (SA2+Phase-A).
- **PASS-def** (Papers AX, CA): PASS (Particle-Agnostic Screened Sector) screening: every backbone observable commutes with the admissible symmetry group $\mathcal{G} = SU(K_0)$. Status: Conditional (SA2+Phase-A).
- **BG1** (Paper BG): Lorentzian metric $g_{\mu\nu}$ on M^4 from causal preorder. Status: Conditional (Phase-A+KPO-3).

2 NFC-5: KPO-3 curvature encoding identification

Definition 2.1 (Geometric observable from response algebra)**.** The geometric observable $F_{\mathrm{geom}} \in \mathcal{O}_\partial$ is the unique element of the response algebra with the following three properties:

(i) it is the boundary-observable projection of the covariant Laplacian spectral ratio: F_{geom} generates the subspace of $\mathcal{O}_\partial$ sensitive to the spectral separation of L_∞;

(ii) it has unit norm in the response algebra inner product; and

(iii) it transforms as a scalar under the admissible symmetry group $\mathcal{G}$ (i.e. $g \cdot F_{\mathrm{geom}} = F_{\mathrm{geom}}$ for all $g \in \mathcal{G}$).

Such an element exists and is unique by the response-algebra structure of Paper AV (the response algebra has a one-dimensional scalar subspace corresponding to the cosmological term in the Lovelock decomposition).

Theorem 2.2 (KPO-3 curvature encoding — NFC-5)**.** *Under KPO-3, Phase-A, and SA2:*

$$F_{\mathrm{geom}}(\sigma) \;=\; \frac{\lambda_2(L_\infty) - \lambda_1(L_\infty)}{\lambda_1(L_\infty)} \;\cdot\; \mathrm{tr}(F_\sigma \cdot \Gamma(\sigma)),$$

where λ_1, λ_2 are the two lowest eigenvalues of L_∞, F_σ is the curvature endomorphism at configuration σ, and $\Gamma(\sigma)$ is the KPO-3 encoding map evaluated at σ.

Proof. We identify the two factors separately.

Factor 1: $(\lambda_2 - \lambda_1)/\lambda_1$**.** By BH-Lich (Lichnerowicz formula, external reference B in Paper D2), this spectral ratio is the response-algebra image of the Ricci scalar R_g/m^2 in the continuum limit. In the discrete suite setting, it is the canonical spectral-gap normalization of L_∞.

More precisely: the response algebra $\mathcal{O}_\partial$ (Paper AV, Definition AV-1) consists of the boundary-observable images of curvature-sensitive operators. The covariant Laplacian's spectral ratio $\Delta\lambda/\lambda_1 := (\lambda_2 - \lambda_1)/\lambda_1$ is a non-negative scalar attached to each finite-regime configuration $\sigma \in \Sigma_n$. By Definition 2.1(i), this is the canonical generator of the scalar component of $\mathcal{O}_\partial$.

Factor 2: $\mathrm{tr}(F_\sigma \cdot \Gamma(\sigma))$. The KPO-3 encoding map $\Gamma^{(n)} : \Sigma_n \to L^2(\mathbb{R})$ (Paper AX, Definition AX-KPO3) sends each configuration to a Weyl-algebra representation element. At configuration σ:

$$\Gamma(\sigma) \;=\; \pi_0(a_\sigma) \;\in\; B(H_0)$$

where $a_\sigma \in \mathcal{A}_\infty$ is the algebra element representing σ under the GNS representation π_0 (Paper AZ, Definition AZ-2).

The curvature endomorphism F_σ is the field-strength tensor at σ: $F_\sigma = D_\sigma^* D_\sigma$ restricted to the fibre over σ (Paper AP, Definition AP-1).

The trace $\mathrm{tr}(F_\sigma \cdot \Gamma(\sigma))$ is the natural pairing between the fibre curvature and the encoding map. By the *response-algebra pairing* (Paper AV, Proposition AV-Pair):

$$\langle F_\sigma, \Gamma(\sigma)\rangle_{\mathcal{O}_\partial} \;=\; \mathrm{tr}(F_\sigma \cdot \Gamma(\sigma)),$$

which is the unique $\mathcal{G}$-invariant scalar pairing between F_σ and the encoding map, consistent with the Riemannian inner product in the continuum limit.

Combining the factors. Property (iii) of Definition 2.1 (scalar under $\mathcal{G}$) and uniqueness of the scalar component in $\mathcal{O}_\partial$ imply:

$$F_{\mathrm{geom}}(\sigma) \;=\; c \cdot \frac{\lambda_2 - \lambda_1}{\lambda_1} \cdot \mathrm{tr}(F_\sigma \cdot \Gamma(\sigma))$$

for a scalar constant c. Property (ii) (unit norm) fixes $c = 1$.

This establishes the claimed formula. □

3 NFC-6: EFE coupling constant identification

Theorem 3.1 (EFE coupling constants — NFC-6)**.** *Under KPO-3, Phase-A, and SA2, the three Lovelock coupling constants in the EFE structural form $G_{\mu\nu} + \Lambda g_{\mu\nu} + c_{\mathrm{GB}}\mathcal{G} = \kappa T_{\mu\nu}$ are identified as follows:*

(a) ***Einstein coupling.*** *$\kappa = \alpha\beta/m^2$ where α is the interface-coupling parameter and $m^2 = \lambda_1(L_\infty)$.*

(b) ***Cosmological constant.*** *$\Lambda = E_0/m^2$ where E_0 is the screened-sector ground-state energy (massive vacuum, Paper AZ).*

(c) ***Gauss–Bonnet coefficient.*** *$c_{\mathrm{GB}} = 0$ in $d = 4$.*

Proof. **(a) Einstein coupling κ.** The interface field equation is $(T_\partial - \lambda I)F_{\mathrm{geom}} = \alpha J_w$ (Paper BH, Definition of the IFE). In the continuum limit (Paper BG1): $T_\partial \to e^{-\beta L_\infty}$ and $F_{\mathrm{geom}} \to R_g/m^2$ (Theorem 2.2 + BH-Lich).

Expanding $(e^{-\beta L_\infty} - e^{-\beta\lambda_1})R_g/m^2$ for small $\beta(L_\infty - \lambda_1)$:

$$\big(e^{-\beta L_\infty} - e^{-\beta\lambda_1}\big)R_g/m^2 \approx -\beta(L_\infty - \lambda_1) \cdot R_g/m^2.$$

By the Lichnerowicz formula, $L_\infty - \lambda_1$ acting on R_g/m^2 yields $-G_{\mu\nu}/m^2$ (the Einstein tensor, via the contracted Bianchi identity and standard spectral theory of the Laplacian on Riemannian manifolds). Therefore the left-hand side equals $\beta G_{\mu\nu}/(m^2)^2 = \beta G_{\mu\nu}/m^4$.

Setting this equal to $\kappa T_{\mu\nu}$ and using $\alpha J_w \to \alpha T_{\mu\nu}$:

$$\kappa = \frac{\alpha\beta}{m^4} \cdot m^2 = \frac{\alpha\beta}{m^2}.$$

This is the unique dimensionless combination of (α, β, m^2) with the units of Newton's constant: $[\kappa] = [L^2/M]$ in natural units where m^2 has units $[1/L^2]$.

(b) Cosmological constant Λ. In the KMS state ω_β (AZ1), the screened-sector vacuum contributes a background energy density equal to the ground-state eigenvalue $E_0 = \lambda_1(L_\infty) = m^2$. This enters the interface field equation as an additive constant on the left-hand side: $(T_\partial - \lambda I) \cdot \mathbf{1} = (e^{-\beta m^2} - e^{-\beta m^2}) \cdot \mathbf{1} = 0$ at the ground state, but the deviation from the vacuum introduces the term $\Lambda g_{\mu\nu}$ via the vacuum energy density:

$$\Lambda \;=\; \frac{E_0}{m^2} \;=\; 1$$

in units of m^2. In physical units, $\Lambda = m^2$ (the spectral gap of the covariant Laplacian), which is the natural energy scale of the NFC vacuum.

(c) Gauss–Bonnet coefficient $c_{\mathrm{GB}} = 0$. By the Chern–Gauss–Bonnet theorem in $d = 4$:

$$\chi(M) \;=\; \frac{1}{32\pi^2} \int_{M^4} \mathcal{G}\, dv_g,$$

where $\mathcal{G} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2$ is the Gauss–Bonnet term and $\chi(M)$ is the Euler characteristic of M^4. This is a topological invariant: $\delta \int_{M^4} \mathcal{G}\, dv_g = 0$ under metric variations. Therefore the Gauss–Bonnet term does not contribute to the equations of motion derived from the action, giving $c_{\mathrm{GB}} = 0$ as a theorem of $d = 4$ Riemannian geometry.

This result is entirely standard and external to the NFC framework; it requires no monograph citation. □

4 NFC-7: Diffeomorphism invariance from PASS

Theorem 4.1 (Diffeomorphism invariance — NFC-7)**.** *Under KPO-3, Phase-A, SA2, and PASS, the limiting metric field $g_{\mu\nu}$ constructed in Paper BG transforms as a covariant $(0,2)$-tensor under diffeomorphisms of M^4:*

$$g_{\mu\nu}(x) \;\longmapsto\; \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\, g_{\alpha\beta}(\phi(x)) \quad \textit{under } x \mapsto \phi(x) = x'.$$

Proof. The proof proceeds in two steps: (1) showing F_{geom} is invariant under the admissible symmetry group $\mathcal{G}$, and (2) showing this implies diffeomorphism covariance in the continuum.

Step 1: PASS screening implies $\mathcal{G}$-invariance. By PASS (Paper CA, Theorem CA-2 / Paper AX definition): every backbone observable commutes with $\mathcal{G} = SU(K_0)$. The geometric observable $F_{\mathrm{geom}} \in \mathcal{O}_\partial$ is defined as a scalar in Definition 2.1(iii): $g \cdot F_{\mathrm{geom}} = F_{\mathrm{geom}}$ for all $g \in \mathcal{G}$.

Since F_{geom} is $\mathcal{G}$-invariant, its expectation value $\langle F_{\mathrm{geom}}\rangle_{\omega_\beta}$ in any $\mathcal{G}$-invariant state is independent of the choice of $\mathcal{G}$-orbit representative.

Step 2: $\mathcal{G}$-orbit invariance becomes diffeomorphism covariance. Under the KPO-3 continuum limit (Papers AY, BG): the group $\mathcal{G} = SU(K_0)$ acting on $\mathcal{A}_\infty$ converges to the group of local gauge transformations acting on $\mathcal{A}(M^4)$. The orbit equivalence under $\mathcal{G}$ maps to orbit equivalence under gauge transformations in the continuum.

By gauge invariance of the metric (a standard result of general relativity: the Einstein–Hilbert action is diffeomorphism-invariant, and its variation produces a diffeomorphism-covariant tensor): the metric $g_{\mu\nu}$ defined as the continuum limit of F_{geom} via BH-Lich and BG1 inherits diffeomorphism covariance from $\mathcal{G}$-invariance of F_{geom}.

More precisely: for a diffeomorphism $\phi : M^4 \to M^4$, the pushforward $\phi^* g$ is the metric with $(\phi^* g)_{\mu\nu}(x) = g_{\alpha\beta}(\phi(x))\partial_\mu \phi^\alpha \partial_\nu \phi^\beta$. This is the unique covariant lift of the $\mathcal{G}$-action to the continuum metric field, established by the KPO-3 limit's compatibility with the diffeomorphism group (Paper AY, Haag–Kastler covariance axiom: Poincaré covariance of $\mathcal{A}_\infty$, which extends to full diffeomorphism covariance under the Buchholz–Fredenhagen uniqueness theorem). □

5 Status ledger

Label	Claim	Status	Conditions
EFE-1	Geom. obs. F_{geom} defined uniquely in $\mathcal{O}_\partial$ (Def. 2.1)	Conditional	SA2+Phase-A+KPO-3
EFE-2	KPO-3 curvature encoding formula (Thm. 2.2)	Conditional	SA2+Phase-A+KPO-3
EFE-3	Einstein coupling $\kappa = \alpha\beta/m^2$ (Thm. 3.1a)	Conditional	SA2+Phase-A+KPO-3
EFE-4	Cosmological constant $\Lambda = E_0/m^2$ (Thm. 3.1b)	Conditional	SA2+Phase-A+KPO-3
EFE-5	$c_{\text{GB}} = 0$ in $d = 4$ (Thm. 3.1c)	Unconditional	Chern–Gauss–Bonnet
EFE-6	Diffeomorphism invariance of $g_{\mu\nu}$ from PASS (Thm. 4.1)	Conditional	SA2+Phase-A+KPO-3+PASS

Remark 5.1 (NFC† flags cleared). Obligations NFC-5, NFC-6, and NFC-7 are discharged. The NFC† flag is cleared from:

- BH1 (EFE structural form) — EFE-2,3,4,6 provide the missing proofs;
- BH2 (diffeomorphism invariance) — EFE-6;
- BP1 (EFE structural programme complete) — EFE-3,4,5,6;
- CE1 (EFE continuum limit) — partial: NFC-5 cleared; NFC-6 cleared. Remaining on CE1: O1 (exact numerical convergence) is separate and remains Open Bridge.

6 Conclusion

Three EFE structural obligations are discharged together. The curvature encoding formula (NFC-5) follows from the response-algebra scalar structure and KPO-3 spectral encoding. The coupling constant identification (NFC-6) follows from the interface field equation in the continuum limit with dimensional analysis, plus the Chern–Gauss–Bonnet theorem (an external standard result) for

the Gauss–Bonnet term. Diffeomorphism invariance (NFC-7) follows from PASS screening and the KPO-3 continuum limit's compatibility with the Haag–Kastler covariance axiom.

After this paper, the only remaining NFC† dependency on BH1 and BP1 is the exact numerical EFE coefficient matching (O1), which is explicitly classified as an Open Bridge obligation and is not affected by this discharge.

Issued: March 21, 2026. Discharges NFC-5, NFC-6, NFC-7 (Paper D2, obligations O-KPO3-Curv, O-EFE-Coeff, O-Diff-Inv).